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Internet-based remote exercise interventions for maintenance hemodialysis patients: a systematic review of the evidence

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Abstract

Objective This study aimed to synthesize the relevant evidence regarding remote exercise management for patients undergoing Maintenance hemodialysis(MHD) using an evidence-based approach. The goal was to provide clinicians with evidence-based recommendations to facilitate the more effective application of internet technologies in managing remote exercise for this patient population.

Methods A comprehensive search was conducted across multiple databases, professional association websites, guideline repositories, and evidence-based resource platforms to identify all relevant evidence on exercise in MHD patients. The search covered publications from database inception to January 31, 2025. Two independent reviewers evaluated the quality of eligible studies, extracted data according to the inclusion criteria, and assigned evidence levels.

Results A total of 18 publications were included: 2 guidelines, 2 expert consensuses, 4 systematic reviews, 9 randomized controlled trials, and 1 quasi-experimental study. Thirty-two best evidence statements were synthesized across seven domains: remote exercise preparation, remote exercise requirements, pre-exercise assessment, exercise prescription, remote exercise monitoring, exercise support, and precautions.

Conclusion The evidence synthesized in this study provides preliminary guidance for the development of remote exercise management strategies in MHD patients. It may assist healthcare professionals in raising awareness of exercise management and in exploring the integration of internet technologies to improve support for this population. At the same time, further high-quality, multicentre studies are needed to confirm these findings and establish standardized approaches for broader clinical application.

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Keywords Internet, Maintenance hemodialysis, Remote, Exercise management, Best evidence synthesis

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Introduction

Maintenance hemodialysis (MHD) stands as the most prevalent life-prolonging treatment for patients with End-stage Renal Disease (ESRD) [1]. As of 2023, the global number of patients with Kidney Failure with Replacement Therapy (KFRT) has reached approximately 4.59 million, and this figure is projected to continue rising [2]. Although MHD improves survival, patients frequently experience a profound decline in physical function, exercise capacity, and Health-related Quality of Life (QoL), largely attributable to sedentary behavior, muscle wasting, and dialysis-related fatigue. These factors contribute to increased morbidity, mortality, and healthcare utilization [2, 3].

Recognizing this, traditional exercise interventions, including intradialytic exercise (IDE) and supervised home-based exercise (HBE), have been implemented and demonstrate positive effects on physical performance and QoL in this population [4, 5]. However, the widespread adoption of these conventional approaches faces significant practical barriers. Patient adherence is often hampered by post-dialysis fatigue, logistical constraints (e.g., transportation, timing), and a lack of accessible, supervised programs. Furthermore, resource limitations within clinics, including insufficient staff and space, restrict the scalability of structured IDE programs [6].

Internet-based remote exercise interventions have emerged as a promising alternative to address practical and logistical barriers of intradialytic and conventional home-based exercise programs. These interventions leverage digital platforms — including mobile health applications, video-conferencing, wearable activity trackers, and interactive web or virtual-reality modules — to deliver individualized exercise prescriptions, real-time physiological and behavioral feedback, and remote behavioural support (e.g. automated reminders, goal setting, clinician messaging). Recent systematic reviews and primary trials indicate that digital interventions can increase physical activity and improve patient-reported outcomes in people receiving dialysis, and that mobile/telehealth delivery can achieve comparable short-term improvements in functional capacity and QoL versus traditional approaches. Importantly, digital delivery enables continuous monitoring (heart rate, step counts, session logs), remote safety checks, and data capture for adherence analytics — capabilities that are difficult to achieve with unsupervised home exercise or limited in-center intradialytic programs. These features may support better long-term engagement and scalable delivery across dispersed populations, although the current evidence base is preliminary and heterogeneous [7, 8].

Despite this potential, the evidence base for internet-based remote exercise remains nascent and distinct from the broader literature on traditional exercise. Existing

systematic reviews have primarily synthesized evidence for general or IDE, often with limited focus on the unique mechanisms, implementation strategies, and outcomes associated with digitally delivered interventions. Consequently, the current evidence is characterized by a prevalence of small-scale, single-centre randomized controlled trials (RCTs) and expert consensus statements, resulting in findings that are often of low-to-moderate quality and lack standardization [9].

Therefore, this systematic review aims to synthesize the available evidence pertaining specifically to internet-based remote exercise interventions for MHD patients. Through a critical appraisal of the methodological quality of the included studies, this review seeks to offer valuable insights for clinical practice and contribute to the foundation for further research in this evolving area.

Materials and methods

Study protocol

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. The review protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO ID: CRD420251037918).

Sources of information

We conducted a comprehensive literature search across multiple sources to identify relevant evidence on internet-based remote exercise management for MHD patients. Global guideline repositories and nephrology professional association websites were searched, including the International Guideline Network, Medlive.cn, Registered Nurses' Association of Ontario, National Institute for Health and Care Excellence, National Guideline Clearinghouse, National Health and Medical Research Council, and the World Health Organization. Additional nephrology-specific professional association websites were also consulted, such as the Chinese Society of Nephrology, the Renal Association, and the National Kidney Foundation. Evidence-based resource databases were searched, including BMJ Best Practice, UpToDate, the Joanna Briggs Institute (JBI) EBP Database, and the Cochrane Library. Furthermore, bibliographic databases were searched, covering both international and Chinese databases: PubMed, Embase, Web of Science, Ovid, CINAHL, TRIP Database, China National Knowledge Infrastructure (CNKI), Chinese Biomedical Literature Database (CBM), Chinese Science and Technology Journal Database (VIP), and Wanfang Data Knowledge Service Platform (Wanfang Data).

Search strategy

The search covered the period from database inception to January 31, 2025. Search terms combined controlled vocabulary (e.g., MeSH/Emtree/CINAHL Headings) and free-text keywords related to “maintenance hemodialysis,” “exercise,” and “internet-based or remote interventions.” The search strategy was adapted to the syntax and indexing terms of each database. No language restrictions were applied. A detailed example of the full search strategies for each database is provided in Additional file 1.

The English search terms for guideline websites, professional association websites, and evidence-based resource repositories included: Renal Dialysis, Kidney Failure, Chronic, Renal Insufficiency, Chronic, hemodialysis, haemodialysis, hemo-dialysis, haemo-dialysis, HD, maintenance hemodialysis, maintenance haemodialysis, maintenance HD, MHD, end stage renal disease, end-stage renal disease, ESRD, end stage kidney disease, end-stage kidney disease, ESKD, chronic kidney disease, CKD, CKD 5, CKD5, CKD stage 5, uremia, uraemia, uremic, uraemic, dialysis, kidney replacement therapy, KRT, dialysis patient, dialysis patients, hemodialysis patient, hemodialysis patients; Exercise, Exercise Therapy, Telemedicine, Remote Consultation, Mobile Applications, Internet-Based Interventions, Wearable Electronic Devices, Video Games, Social Media, Text Messaging, Videoconferencing, Monitoring Ambulatory, Artificial Intelligence, tele-exercise, teleexercise, remote exercise, distance exercise, internet-based exercise, web-based exercise, online exercise, physical activity, home-based exercise, home exercise, training, home-based training, Movement, mHealth, m-health, eHealth, e-health, digital health, telehealth, Habilitation, telerehabilitation, tele-rehabilitation, remote rehabilitation, app, apps, application, applications, smartphone app*, mobile app*, phone app*, health app*, wearable, wearables, wearable device*, wearable technology, fitness tracker*, activity tracker*, smartwatch, smart watch, accelerometer, biofeedback, virtual training, virtual rehabilitation, virtual reality, VR, augmented reality, AR, exergam*, serious gam*, video game*, remote consultation, remote monitoring, tele-monitoring, tele-monit, ing, tele-monitoring, tele-monit, ing, videoconferenc*, video conferenc*, teleconferenc*, text message*, SMS, chatbot*, social media.

Study eligibility criteria

Study inclusion was predicated on a predefined Population, Intervention, Comparison, Outcomes, and Study Design (PICOS) framework [10].

Population(P): Adult patients undergoing MHD.

Intervention(I): Internet-based or remote exercise interventions (e.g., tele-exercise, web-based programs,

mobile APP, wearable devices, telehealth or telerehabilitation platforms).

Comparison(C): Usual care, conventional in-center exercise programs, other non-digital interventions, or no intervention.

Outcomes(O): Studies reporting at least one of the following—exercise adherence, physical function (e.g., 6-minute walk distance(6MWT), muscle strength), quality of life, dialysis-related symptoms (e.g., fatigue, sleep quality), cardiovascular or metabolic outcomes, or safety/adverse events.

Study design(S): Published guidelines, expert consensus, recommended practices, evidence syntheses, systematic reviews, RCTs, and quasi-experimental studies. No restrictions regarding language were made.

The exclusion criteria were: (1)Non-human studies, pediatric populations, or studies not involving MHD patients. (2)Interventions unrelated to exercise or lacking an internet-based/remote delivery component. (3) Conference abstracts, editorials, commentaries, letters, or protocols without primary data. (4)Duplicate publications of the same study (in such cases, the most complete or latest version was included). (5) Studies with insufficient data to extract relevant outcomes.

Studies selection and data extraction

Two researchers independently screened and evaluated the quality of the literature strictly following the inclusion and exclusion criteria, beginning with titles and abstracts. Blinding to author names, institutions, or journal information was not implemented at any stage. Full texts were subsequently reviewed for further screening. Any discrepancies arising during the selection process were first discussed between the two reviewers; if consensus could not be reached, a third reviewer was consulted to adjudicate and finalize the decision.

A standardized data extraction form was then used to systematically organize the findings. Extracted information included: author(s), type of literature, country of publication, publication date, journal name, relevant research content, and specific measures. During data extraction, blinding was likewise not applied to author, institution, or journal information. Two researchers independently extracted the data and cross-checked each other's results. Any disagreements were resolved through discussion, with unresolved issues referred to a third reviewer to ensure accuracy and consistency.

Risk of bias and quality assessment

Two reviewers independently assessed the methodological quality and risk of bias of all included studies using validated tools tailored to each study design. Any disagreements were resolved through discussion or consultation with a third reviewer, in line with the PRISMA

2020 guidelines and Cochrane Handbook recommendations. Specifically:

- 1) Clinical practice guidelines and expert consensus statements were evaluated using the updated Appraisal of Guidelines for Research and Evaluation II (AGREE II, 2017) instrument.
- 2) Systematic reviews were appraised with A Measurement Tool to Assess Systematic Reviews 2 (AMSTAR 2).
- 3) Randomized controlled trials (RCTs) were assessed using the Cochrane Risk of Bias 2.0 tool (RoB 2).
- 4) Quasi-experimental studies were evaluated with the Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) tool.

This multi-tiered assessment approach ensured a rigorous and comprehensive evaluation of methodological quality across diverse study designs, providing a robust basis for synthesizing the evidence [11–13].

Evidence classification and grading process

Given the inclusion of heterogeneous evidence types—including systematic reviews, RCTs, quasi-experimental studies, clinical practice guidelines, and expert consensus statements—a stratified and weighted approach was adopted to ensure methodological rigor and comparability across evidence sources. Two reviewers independently conducted the assessments and completed data extraction using a standardized form. During quality appraisal, the reviewers were not blinded to authorship or journal information. Any discrepancies were resolved through discussion or consultation with a third reviewer to ensure consensus. After methodological assessment, the certainty of evidence supporting each recommendation was evaluated and weighted using the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) framework. The GRADE approach considered study limitations, consistency, directness, precision, and publication bias to determine the overall confidence in the evidence (high, moderate, low, or very low). Evidence derived from systematic reviews and high-quality RCTs was assigned greater weight, while quasi-experimental studies, guidelines, and expert consensus statements provided supplementary support where experimental evidence was limited. Finally, all recommendations were synthesized and stratified according to their corresponding evidence level and GRADE rating. This stratification and weighting approach allowed clear differentiation between evidence supported by robust empirical data and that based primarily on expert or contextual consensus, thereby enhancing the transparency, interpretability, and clinical credibility of the final recommendations [14].

Result

General characteristics of the included literature

A total of 1,564 articles were initially searched. After removing duplicates using EndNote X9 software, 436 articles remained. After reviewing titles and abstracts, 358 articles were excluded based on the inclusion criteria, resulting in 78 articles. Following a full-text review, 18 articles were finally included. These comprised 2 guidelines, 2 expert consensus statements, 4 systematic reviews, 9 RCTs, and 1 quasi-experimental study. The literature search flowchart is shown in Fig. 1. The basic characteristics and the extracted data of the included literatures are provided in Additional file 2. Detailed results of the quality assessment for the included studies are provided in Additional file 3.

Quality assessment results of the included guidelines

Both Baker et al. [15] and Thompson et al. [16] achieved high scores across all AGREE II domains, indicating strong methodological rigor and clinical applicability. These guidelines are therefore considered robust references to guide practice.

Quality assessment results of the included expert consensus statements

Among the 2 expert consensus statements, Battaglia et al. [18] demonstrated high methodological quality, with recommendations grounded in systematic evidence and strong applicability. In contrast, Zhang et al. [17] was rated moderate, offering clinically relevant insights but relying more heavily on expert opinion with limited methodological rigor. Overall, while both statements provide useful guidance, their strength of evidence and reliability differ.

Quality assessment results of the included systematic reviews

Of the 4 included systematic reviews, three (Bernier-Jean et al., [19]; Clyne et al., [20]; Warrington et al., [21]) demonstrated high overall confidence, supported by comprehensive searches, systematic risk of bias assessments, and transparent synthesis methods. In contrast, Hamet et al., 2017 [22] was rated moderate, providing valuable insights but with limitations in protocol registration, search strategy transparency, and bias assessment.

Quality assessment results of the included RCTs

The risk of bias assessment for the 9 included RCTs [23–30] is shown in Figs. 2 and 3. Overall, 5 studies [23–26] were judged to have some concerns regarding risk of bias, primarily due to the inability to blind participants and personnel. 4 studies [27–30] were judged to be at high risk of bias due to issues such as inadequate reporting of allocation concealment. The RCT evidence is generally

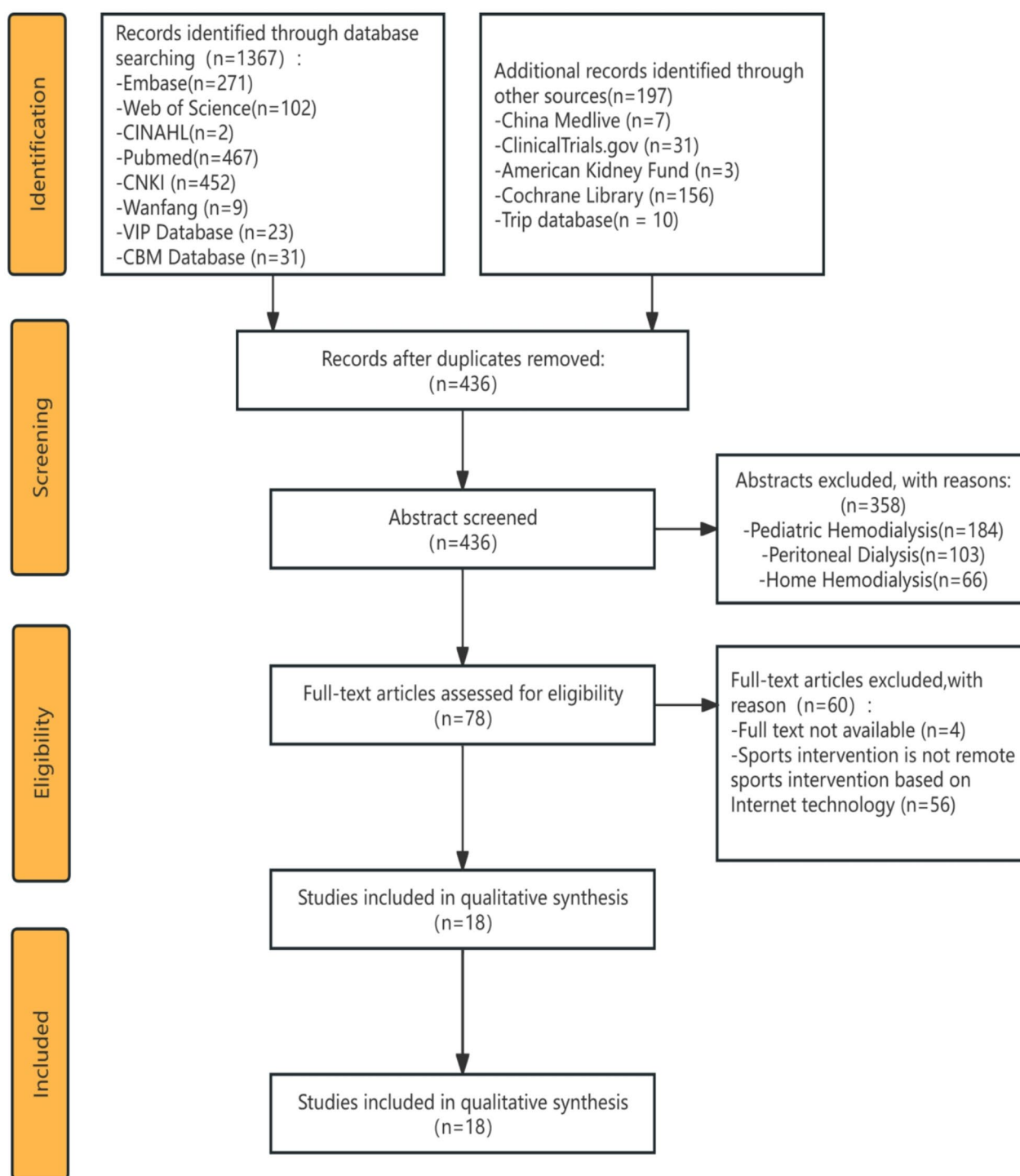


Fig. 1 Flowchart of the literature screening and search process

credible (many trials with low risk in core domains), but the predominance of pilot/small trials with implementation-related limitations means findings should be interpreted with some caution.

Quality assessment results of the included quasi-experimental studies

The quasi-experimental study by Li et al. [31] was rated moderate quality, primarily due to potential residual confounding and deviations from intended interventions inherent to its non-randomized pilot design. Despite

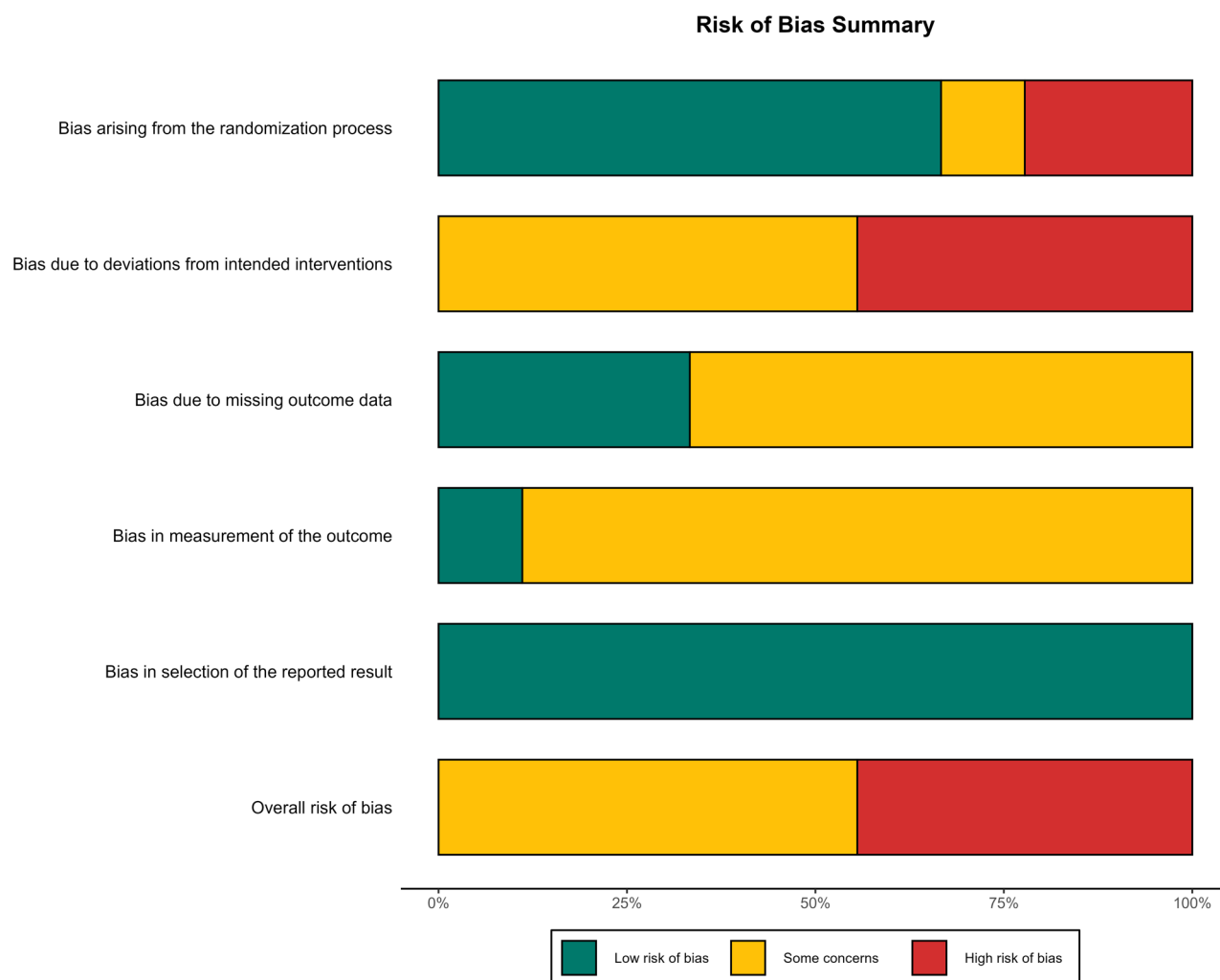


Fig. 2 Risk of bias summary. The bar chart displays the distribution of risk of bias judgments (low risk, some concerns, high risk) for each domain of the Cochrane RoB 2 tool across all included studies

these limitations, the study contributes meaningful feasibility and preliminary outcome data, though its findings are less conclusive compared with randomized evidence.

Overall quality assessment and implications

Overall, the methodological quality of the 18 included studies was moderate, providing a generally credible evidence base for remote exercise management in MHD patients. The guidelines and most systematic reviews demonstrated strong methodological rigor and clinical applicability, while the randomized controlled trials demonstrated moderate-to-low overall quality with identifiable limitations in design and reporting. The expert consensus statements and the quasi-experimental study provided supportive evidence but were constrained by their methodological limitations. Collectively, the existing evidence supports the safety, feasibility, and clinical benefits of exercise interventions for MHD patients, though further large-scale, high-quality studies are

needed to strengthen the reliability and generalizability of these findings.

Summary and analysis of best evidence

Through a process of discussion and analysis of the extracted evidence, the entire body of evidence was integrated. In conclusion, 32 pieces of evidence were synthesised across seven aspects pertinent to MHD patients: remote exercise preparation, remote exercise requirements, Pre-exercise Assessment, exercise prescription, remote exercise monitoring, exercise support, and precautions. The synthesis of the evidence is presented in Table 1. We applied the GRADE framework to the outcomes and the full GRADE evidence profiles (with domain-level reasons for downgrading/upgrading) are supplied in Additional file 4.

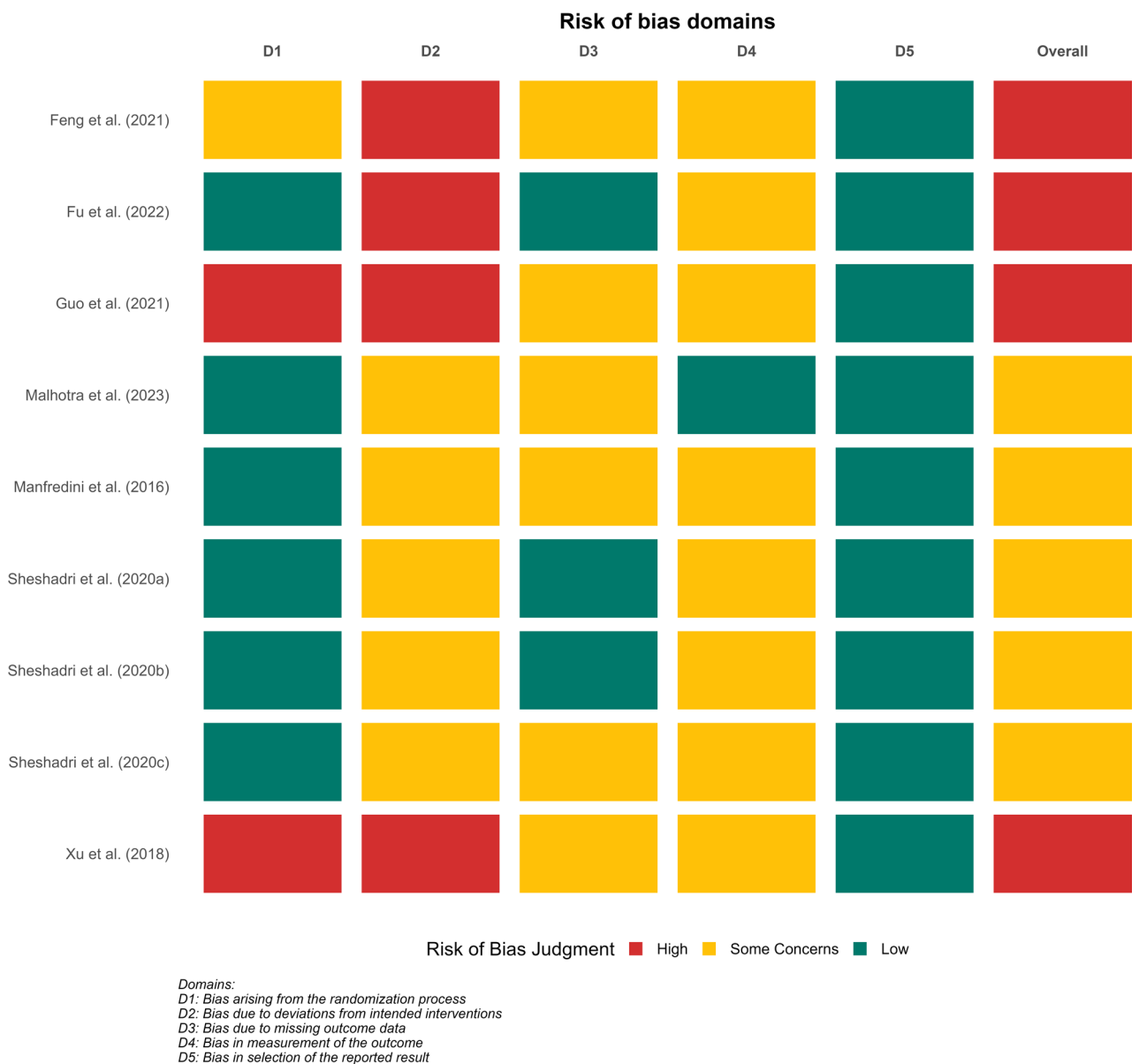


Fig. 3 Risk of bias traffic light plot. Each row represents an individual study, and each column represents a specific risk of bias domain from the Cochrane RoB 2 tool. Green, yellow, and red indicate low risk, some concerns, and high risk of bias, respectively

Discussion

Overview of findings

This systematic evidence synthesis summarized current research on internet-based remote exercise management for MHD patients. Eighteen studies were included, encompassing guidelines, expert consensus statements, systematic reviews, RCTs, and quasi-experimental study. Through structured synthesis and quality weighting, 32 recommendations were formulated across seven domains: preparation, exercise requirements, pre-exercise assessment, exercise prescription, remote monitoring, exercise support, and safety precautions. These recommendations provide a structured reference

framework for clinicians to guide remote exercise management among MHD patients.

However, the evidence base is characterized by methodological variability and limited high-quality randomized data. The majority of studies were small-scale or single-centre, often with moderate to high risk of bias, emphasizing that the findings should be interpreted with caution. The present results therefore represent preliminary guidance, forming the basis for future high-quality trials and implementation research.

Table 1 Summary of the best evidence for remote movement management of maintenance Hemodialysis patients based on internet technology

Evidence items	Evidence content	GRADE	Recommendation
Remote exercise preparation	1. Establish a multidisciplinary exercise management team (nephrology, nursing, rehab, cardiology, psychology) [15–18]	Moderate	Strong
	2. Provide regular training for team members on remote exercise protocols and devices [15–20]	High	Strong
	3. Screen patients for exercise contraindications before enrolment [17–22]	Moderate	Strong
Remote exercise requirements	4. Ensure safe exercise venue per prescription (home/community/gym) [15–18, 31]	Moderate	Conditional
	5. Obtain informed consent for remote exercise (risks, benefits, data use) [15–32]	High	Strong
	6. Provide concise exercise demonstrations (manuals, video, live demos) [17–25]	High	Strong
Pre-exercise Assessment	7. Use remote assistance tools: phone, APP, WeChat, wearables, platforms. [15, 22–32]	High–Moderate	Strong
	8. Confirm telemedicine devices/apps are properly installed and safe [21–22, 28]	Moderate	Conditional
	9. Collect baseline demographics and clinical data (age, dialysis vintage, comorbidity) [17–20]	Moderate	Strong
	10. Assess exercise capacity via Graded exercise test (GXT) or validated field tests (6MWT, the 30s sit-to-stand test and timed get up and go test (TUG)) [17–19, 26]	High	Strong
	11. Record baseline physical activity level [15, 17–25]	High	Strong
	12. Record patient preferences for exercise modality [23–32]	Moderate	Strong
Exercise prescription	13. Conduct physical exam and anthropometrics (BMI, strength) [15–20]	Moderate	Strong
	14. Frequency (F): prescribe aerobic 3–5x/wk; resistance 3–5x/wk; flexibility ≥ 3x/wk [15–19, 23–32]	Moderate–High	Strong
	15. Intensity (I): moderate aerobic (RPE 11–13 or 50–70% HRmax); resistance 60–70% 1RM [18–29]	Moderate	Strong
	16. Type (T): aerobic (walking, cycling), resistance (bands, weights), flexibility (tai chi, Baduanjin) [15–19, 23–32]	Moderate–High	Strong
	17. Time (T): aerobic 20–60 min/session; resistance ~ 20–30 min/session [15–19]	Moderate	Strong
	18. Real-time monitoring of vitals (pre/during/post) via telemedicine devices [22–32]	Moderate–High	Strong
Remote exercise monitoring	19. Heart rate monitoring via wearables / patient self-monitoring [21–25]	Moderate	Strong
	20. Home BP monitoring education (non-fistula arm) [18–22]	Moderate	Strong
	21. SpO ₂ monitoring with portable pulse oximeter as indicated [17–22]	Moderate	Conditional
	22. Maintain exercise log (duration/intensity/vitals/discomfort) [17–22, 25–32]	Moderate–High	Strong
	23. Define clear exercise termination criteria and escalation pathways [15–18]	High	Strong
	24. Provide patient/family education on benefits and adherence strategies [15–26]	Moderate	Strong
Exercise support	25. Implement structured follow-up to monitor adherence and goals [19–28]	High	Strong
	26. Set realistic, individualized exercise goals with multidisciplinary team input [15–18, 23–27]	Moderate	Strong
	27. Use digital “check-in” or punching systems (APP/platform) to track completion. [23–32]	Moderate	Strong
	28. Maintain remote contact (apps/communication tools) to address queries [23–32]	High	Strong
	29. Combine incentives/social support, peer support, goal-setting [24–32]	High	Strong
	30. Reassess and adjust exercise prescription every 3–6 months [15–19]	High–Moderate	Strong
Precautions	31. Warm-up (5–10 min) and cool-down (5–10 min) around sessions [15–18, 23–31]	Moderate	Strong
	32. Protect vascular access during exercise (avoid fistula arm stress) [15–18, 22–32]	Moderate	Strong

Heterogeneity of study design and methodological quality

Substantial heterogeneity exists among the included studies. The differences are reflected in study design, intervention type, participant characteristics, exercise modalities, and outcome measurements. Specifically, interventions ranged from pedometer- [9–24] and wearable device–assisted walking programs [23, 26] to smartphone app–based exercise monitoring [27, 29], social media–enabled group exercise [25, 30, 31], and VR

applications [28]. Outcome indicators also varied, including physical function tests (6MWT, 30-s chair stand, handgrip strength), dialysis adequacy (spKt/V), activity level (step counts or Physical Activity Scale), self-efficacy, adherence, and quality of life.

Because of this heterogeneity, the synthesis was conducted using a stratified, weighted evidence approach rather than quantitative pooling. The methodological rigor of included studies was uneven: while the guidelines

and three of four systematic reviews demonstrated high methodological quality, most RCTs and quasi-experimental studies exhibited potential biases, particularly regarding allocation concealment and blinding. Therefore, the conclusions drawn are indicative rather than confirmatory.

Stratification, weighting, and certainty of evidence

To ensure transparency and rigor, the included evidence was evaluated using validated appraisal tools: AGREE II for guidelines [15, 16] and expert consensus statements [17, 18], AMSTAR 2 for systematic reviews [19–22], RoB 2.0 for RCTs [23–30], and ROBINS-I for quasi-experimental study [30]. Each evidence source was weighted according to methodological quality and direct relevance to remote management. Subsequently, certainty of evidence was graded using the GRADE framework.

Regional and technological variations

Marked regional differences were observed in the design and implementation of remote exercise programs. In Asia, particularly China, interventions frequently incorporated social platforms such as WeChat, mobile apps for home-based guidance, and culturally tailored modalities like Baduanjin or VR-assisted rehabilitation. In Western countries, research more commonly employed wearable devices, pedometer-based walking programs, and telehealth-supported coaching. These variations reflect differences in healthcare infrastructure, cultural exercise habits, and technology availability [19–22].

Such diversity has implications for generalizability. While social media-based group interventions may be highly effective in regions with widespread smartphone access, wearable-based interventions may be more suitable in systems with integrated electronic health platforms. Thus, program design should align with local technological resources, patient literacy, and healthcare delivery models [31].

Remote exercise intervention modalities and evidence synthesis

This review synthesizes distinct modalities of internet-based remote exercise interventions for MHD patients, classifying them into four primary types based on varying evidence levels. While most recommendations align with general exercise guidelines, several are specific to remote management, including the use of wearables, smartphone-based monitoring, digital check-ins, real-time vital tracking, cloud-based data systems, and virtual coaching platforms—all shown to enhance engagement, adherence, and safety:

- 1) Tele-supervised Exercise: This involves real-time, synchronous supervision of exercise sessions by

a healthcare professional via video-conferencing tools. The evidence for this is growing but is currently based on smaller, single-center trials. These studies typically configure exercise around traditional aerobic and resistance training, with the therapist providing immediate form correction and motivation.

- 2) App-/Platform-Mediated Programs: This category, which has the largest and most robust evidence base from recent studies, utilizes dedicated mobile applications or web platforms to deliver pre-recorded exercise videos, personalized workout plans, and educational content. Configuration is often based on the Frequency, Intensity, Type, and Time (FITT) principle, with exercises adapted for the home environment using minimal equipment.
- 3) Wearable Device-Guided Interventions: These interventions, also well-supported by evidence, center on the use of activity trackers (pedometers, accelerometers) to monitor volume (e.g., daily steps) and intensity (e.g., heart rate). The exercise is configured as a goal-setting program, where patients are prescribed daily or weekly step counts or active minutes, with feedback provided through the device or linked app.
- 4) VR and Exergaming: This represents an emerging modality with promising but preliminary evidence. Current studies configure exercise as immersive VR games or somatosensory games that require physical movement, aiming to enhance enjoyment and adherence.

Notably, APP-/platform-mediated and wearable-guided interventions currently hold the strongest evidence [15, 23–26]. However, most studies treat remote exercise as an extension of conventional programs rather than a distinct model, highlighting a critical evidence gap. Future research should directly compare remote and in-person exercise to determine relative effectiveness, feasibility, and cost-effectiveness, while advancing hybrid models that integrate data from wearables with adaptive coaching platforms [27].

Barriers to technology adoption and implementation feasibility

Despite promising outcomes, significant challenges remain regarding the widespread implementation of internet-based exercise management. Barriers include digital illiteracy among older adults, limited access to smart devices, unstable internet connectivity, privacy and data security concerns, and the absence of standardized digital platforms within clinical workflows. These factors can hinder participation and long-term adherence [21, 22, 26].

Addressing these barriers requires tailored solutions such as providing patient education, simplifying user interfaces, offering technical assistance, and ensuring compatibility with hospital information systems. Moreover, multidisciplinary collaboration involving nurses, physiotherapists, and nephrologists is essential to sustain program quality and safety in remote settings [15–18].

Methodological limitations and need for robust evidence

Although the included RCTs demonstrated the potential benefits of internet-based exercise, methodological weaknesses limit confidence in these findings. Blinding of participants and personnel was typically infeasible, which increased performance bias [27–30]. Allocation concealment and attrition management were inconsistently reported. In addition, many trials had small sample sizes and short follow-up periods, limiting statistical power and assessment of long-term effects [23, 24].

Given these limitations, the evidence certainty for most clinical outcomes—such as improvements in exercise capacity, dialysis adequacy, and quality of life—was rated as moderate under the GRADE framework. Thus, these findings should be interpreted as preliminary and hypothesis-generating rather than conclusive.

Safety considerations in remote exercise management

Safety remains the cornerstone of exercise management for MHD patients [15, 16]. Across the reviewed studies, few adverse events were reported, suggesting that supervised or remotely monitored exercise is generally safe when appropriately designed [17, 18]. However, consistent adverse event reporting and risk stratification were lacking. Future programs should incorporate standardized monitoring procedures, including real-time tracking of heart rate, blood pressure, and oxygen saturation. Clear protocols for symptom reporting and clinician intervention should also be established to ensure patient safety [26, 30].

The imperative for continuous support and integration into clinical care

The transition from in-person to remote exercise management does not diminish the necessity for sustained professional support; it redefines its delivery [19–22]. The evidence strongly suggests that long-term adherence is unlikely to be achieved through technology alone. Instead, technology serves as a tool to enable a more continuous and data-informed support model [32]. This review synthesizes evidence on key support strategies delivered remotely, including regular motivational messaging, virtual peer support groups, remote goal-setting and review sessions, and the use of digital exercise diaries. The development of a “continuous support plan” is therefore a core component of remote management,

integrating these psychosocial and behavioral strategies with the technological platform. For successful integration into clinical practice, roles and responsibilities must be clearly defined. Dialysis unit nurses, physiotherapists, or dedicated “e-health” coordinators would be responsible for remotely monitoring patient-generated health data, providing timely feedback, and escalating concerns. This model represents a significant shift from episodic, clinic-centric care to a continuous, patient-centered partnership, and its implementation requires strategic planning, training, and potentially new reimbursement structures [31].

Clinical implications

Despite the limitations, this review demonstrates that internet-based remote exercise management offers a promising complementary strategy for MHD patients. Structured remote programs that integrate wearable or smartphone technologies, personalized exercise prescriptions, and continuous follow-up may help enhance physical function, adherence, and quality of life.

However, these recommendations are best regarded as preliminary guidance rather than definitive clinical standards. Implementation should be gradual, prioritizing interventions supported by moderate to high certainty of evidence and adapted to the resources and digital literacy of local populations. Continuous evaluation and iterative refinement of these programs are crucial to ensure safety and sustainability.

Future research directions

Future studies should focus on conducting multicentre, adequately powered RCTs with standardized outcome measures and longer follow-up periods. Comparative studies evaluating different remote exercise modalities—such as wearable-based versus app-based versus VR-assisted programs—are needed to identify the most effective approaches. Cost-effectiveness analyses, equity assessments, and qualitative research into user experience and barriers will further support the translation of remote management programs into practice.

In addition, development of unified digital platforms that integrate physiological data, patient education, and clinical feedback will facilitate scalability and cross-institutional collaboration. The inclusion of artificial intelligence-driven decision support may enhance personalization and early risk detection.

Strengths and limitations

This study represents one of the first comprehensive attempts to synthesize heterogeneous evidence sources on internet-based exercise management for MHD patients. Strengths include a rigorous search strategy, dual independent screening, structured evidence

weighting, and formal application of multiple quality appraisal tools. Moreover, the GRADE framework was adopted to classify evidence certainty, and full data extraction tables are provided in the supplementary materials.

Several limitations must be acknowledged. Although the search strategy imposed no language restrictions, the majority of the 18 included studies were published in Chinese or English, which may introduce language bias and limit the global representativeness of the findings. Publication bias is also a concern, as studies with positive results are more likely to be published, potentially skewing the synthesized outcomes. Significant heterogeneity was observed across the included studies in terms of intervention designs, measurement tools, and implementation settings, which precluded quantitative meta-analysis and complicates direct comparison between studies. Furthermore, many of the trials were small-scale and conducted at single centers, with common methodological shortcomings such as lack of blinding and inadequate allocation concealment. Consequently, the overall certainty of the evidence, as assessed by the GRADE framework, was generally moderate. These limitations underscore the preliminary nature of the current evidence and highlight the need for more rigorously designed, multi-center trials with longer follow-up periods to strengthen the reliability and generalizability of future recommendations.

Conclusion

In conclusion, this review provides a preliminary framework and preliminary guidance for implementing internet-based remote exercise management for MHD patients. It synthesizes how established exercise principles can be enacted through digital technologies and highlights the specific modalities and support mechanisms that constitute effective remote care. However, it is crucial to restate that this guidance is tempered by the recognized limitations of the current evidence base, including study heterogeneity, methodological weaknesses in primary studies, and regional biases. The recommendations should therefore be viewed as a foundation for practice and a catalyst for further research, not as definitive standards.

Abbreviations

MHD	Maintenance hemodialysis
ESRD	End-stage Renal Disease
KFRT	Kidney Failure with Replacement Therapy
QoL	Health-related Quality of Life
IDE	Intradialytic exercise
HBE	Home-based exercise
RCTs	Randomized controlled trials
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
JBI	Joanna Briggs Institute
CNKI	China National Knowledge Infrastructure

CBM	Chinese Biomedical Literature Database
VIP	Chinese Science and Technology Journal Database
Wanfang Data	Wanfang Data Knowledge Service Platform
PICOS	Population, Intervention, Comparison, Outcomes, and Study Design
6MWT	6-minute walk distance
AGREE II	Appraisal of Guidelines for Research and Evaluation II
APP	Mobile application
GXT	Graded exercise test
TUG	Timed get up and go test
BMI	Body Mass Index
RPE	Borg Rating of Perceived Exertion
1RM	One Repetition Maximum
FITT	Frequency, Intensity, Type, and Time
VR	Virtual reality
AI	Artificial Intelligence

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12882-025-04611-4>.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4

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Author contributions

The focus of this study was devised by all writers. The chief investigators were ShuTing Xu, DongXiang Wu, ShengRu Tao and Yu Chen, with YuLi Yang, MingMei Peng, Huan Liu, HuiXia Yu, MinZuo Chen, WeiBao Liang as co-researchers. YuLi Yang, MingMei Peng, Huan Liu and MinZuo Chen oversaw the gathering of data. ShuTing Xu, WeiBao Liang and DongXiang Wu oversaw data analysis and the original draft of the manuscript. ShengRu Tao, Yu Chen and HuiXia Yu made critical revisions for essential intellectual content. All authors contributed subject-matter expertise. The final paper was read, revised and approved by all authors. ShuTing Xu is the study's guarantor.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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